2.41 1st Order DEs: Modelling

Worked Solutions

Note for Integrating Factors: We can take the integrating factor to be any non-zero constant multiple of an integrating factor. For example, $e^{\int \frac{1}{x}\,dx} = e^{\ln|x|} = |x|$. On any interval not containing $x = 0$, $|x|$ differs from x only by a constant factor (namely ± 1), so we may take the integrating factor to be x. So throughout these worked solutions, the modulus signs will be dropped when multiplying the differential equation through by the integrating factor.

1 a)

Inflow is constant. $\therefore \frac{dV}{dt} = k$.

1 b)

Leak rate is proportional to the volume of water present. Outflow $= kV$. $\therefore \frac{dV}{dt} = -kV$.

1 c)

Inflow $= 6$. Leak rate is proportional to the volume of water present. Outflow $= kx$. $\therefore \frac{dx}{dt} = 6 - kx$.

1 d)

Rate of change is proportional to $(200 - V)$. $\therefore \frac{dx}{dt} = k(200 - V)$.

1 e)

Cooling at a rate proportional to $(T - 18)$. $\therefore \frac{dT}{dt} = -k(T - 18)$.

1 f)

Inflow $= 0.5t$. Leak rate is proportional to the amount of pollutant present. $\therefore \frac{dP}{dt} = 0.5t - kP$.

2)

$$\text{Driving Force} = (18 - 3t)\text{ N}, \quad \text{Resistive Force} = 4v\text{ N},$$

$$\text{Mass} = 6\text{ kg}, \quad a = \frac{dv}{dt}.$$

$$\text{Using } F = ma: \quad (18 - 3t) - 4v = 6\frac{dv}{dt}$$

$$\text{Or } \frac{dv}{dt} = 3 - \frac{1}{2}t - \frac{2}{3}v \Rightarrow \frac{dv}{dt} + \frac{2}{3}v = 3 - \frac{1}{2}t.$$

3)

$$\text{Pulling Force} = 25\text{ N}, \quad \text{Resistive Force} = (5 + 0.2t)v\text{ N}, \quad \text{Mass} = 10\text{ kg}, \quad a = \frac{dv}{dt}.$$

$$25 - (5 + 0.2t)v = 10\frac{dv}{dt}$$

$$\text{Or } \frac{dv}{dt} = \frac{5}{2} - \frac{5 + 0.2t}{10}v \Rightarrow \frac{dv}{dt} + \frac{25 + t}{50}v = \frac{5}{2}.$$

4)

Taking downward as positive. Force downwards is weight: $80g$ N.

Force upwards is air resistance: $6v$ N,

$$\text{Mass} = 80 \text{ kg}, \quad a = \frac{dv}{dt}.$$

$$80g - 6v = 80\frac{dv}{dt}$$

$$\frac{dv}{dt} = g - \frac{3}{40}v$$

$$\Rightarrow \frac{dv}{dt} + \frac{3}{40}v = g.$$

5 a)

Driving Force $= 6e^{-t}$ N, Resistive Force $= v^2$ N,

$$\text{Mass} = 2 \text{ kg}, \quad a = \frac{dv}{dt}.$$

$$6e^{-t} - v^2 = 2\frac{dv}{dt}$$

$$\frac{dv}{dt} = 3e^{-t} - \frac{1}{2}v^2$$

$$\therefore \frac{dv}{dt} + \frac{1}{2}v^2 = 3e^{-t}.$$

5 b)

The differential equation is not linear because it contains v^2, so the integrating factor method cannot be applied.

6 a)

$$\text{Forward Force} = 8 \text{ N}, \quad \text{Resistive Force} = 2v\frac{dv}{dt} \text{ N}, \quad \text{Mass} = 5 \text{ kg}, \quad a = \frac{dv}{dt}.$$

$$8 - 2v\frac{dv}{dt} = 5\frac{dv}{dt}$$

$$(5 + 2v)\frac{dv}{dt} = 8$$

$$\frac{dv}{dt} = \frac{8}{5 + 2v}.$$

6 b)

The differential equation is not linear because it contains $\frac{8}{5+2v}$ and cannot be rearranged into the form $\frac{dv}{dt} + P(t)v = Q(t)$, so the integrating factor method cannot be applied.

6 c)

$$\text{Use Separation of Variables: } \int (5+2v)\,dv = \int 8\,dt \Rightarrow 5v + v^2 = 8t + c.$$

$$\text{Substitute } v = 0.5,\ t = 0: \quad 5 \times 0.5 + 0.5^2 = 8 \times 0 + c \Rightarrow c = 2.75.$$

$$5v + v^2 = 8t + 2.75$$

$$\Rightarrow (v + 2.5)^2 - 6.25 = 8t + 2.75$$

$$\Rightarrow (v + 2.5)^2 = 8t + 9$$

$$\Rightarrow v + 2.5 = \pm\sqrt{8t + 9}$$

$$\Rightarrow v = -2.5 \pm \sqrt{8t + 9}.$$

As initial velocity is 0.5 m s^{-1},
we must have $v = -2.5 + \sqrt{8t + 9}$.

7 a)

Concentration $= \frac{12}{80} = 0.15$ kg/l.

7 b)

Concentration $= \frac{S}{V}$ kg/l.

7 c)

Concentration $= \frac{5}{200} = 0.025$ kg/l.
Salt leaves at $3 \times 0.025 = 0.075$ kg/min.

8)

Concentration $= \frac{S}{100}$ kg/l. Salt leaves at $4 \times \frac{S}{100} = \frac{S}{25}$ kg/min.

9 a)

Salt enters the tank at $5 \times 0.3 = 1.5$ kg/min.

9 b)

Concentration $= \frac{S}{100}$ kg/l. Salt leaves the tank at $5 \times \frac{S}{100} = \frac{S}{20}$ kg/min.

9 c)

$\frac{dS}{dt} = \text{in} - \text{out}$, so $\frac{dS}{dt} = 1.5 - \frac{S}{20}$.

10)

Salt enters the tank at $2 \times 0.4 = 0.8$ kg/min. Salt leaves the tank at $2 \times \frac{S}{50} = \frac{S}{25}$ kg/min. $\frac{dS}{dt} = \text{in} - \text{out}$, so $\frac{dS}{dt} = 0.8 - \frac{S}{25}$, where $S(0) = 2$.

11)

Net change in volume $= 3 - 5 = -2$, so $V = 100 - 2t$.

12)

Net change in volume $= 7 - 2 = 5$, so $V = 60 + 5t$, where $t \geq 0$.

13)

Net change in volume $= p - q$, so $V = V_0 + (p - q)t$.

14)

No salt enters the tank (pure water) $= 0$ kg/min. Salt leaves the tank at $4 \times \frac{S}{100} = \frac{S}{25}$ kg/min. $\frac{dS}{dt} = \text{in} - \text{out}$, so $\frac{dS}{dt} = -\frac{S}{25}$.

15 a)

Net change in volume $= 4 - 6 = -2$, so $V = 100 - 2t$.

15 b)

Salt leaves the tank at $6 \times \frac{S}{100-2t} = \frac{6S}{100-2t}$ kg/min.

15 c)

No salt enters the tank (pure water) $= 0$ kg/min. Salt leaves the tank at $\frac{6S}{100-2t}$ kg/min. $\frac{dS}{dt} = \text{in} - \text{out}$, so $\frac{dS}{dt} = -\frac{6S}{100-2t}$.

16)

Net change in volume $= 4 - 6 = -2$, so $V = 100 - 2t$, valid for $0 \leq t \leq 50$. Salt enters at $4 \times 0.2 = 0.8$ kg/min. Salt leaves at $6 \times \frac{S}{100-2t} = \frac{6S}{100-2t}$ kg/min. Thus $\frac{dS}{dt} = \text{in} - \text{out} = 0.8 - \frac{6S}{100-2t}$.

17 a)

Net change in volume $= 3 - 2 = 1$, so $V = 40 + t$.

17 b)

Salt enters the tank at $3 \times 0.5 = 1.5$ kg/min. Salt leaves at $2 \times \frac{S}{40+t} = \frac{2S}{40+t}$ kg/min. $\frac{dS}{dt} = \text{in} - \text{out}$, so $\frac{dS}{dt} = 1.5 - \frac{2S}{40+t}$.

17 c)

The incoming concentration of salt in the water is 0.5 kg per litre. Over time, the tank becomes dominated by the inflow mixture, so the concentration tends towards 0.5 kg per litre.

18 a)

The number 18 represents room temperature, so 18°C.

18 b)

$(T - 18)$ is the difference between the tea's temperature and 18°C, so it represents how far the tea is above room temperature.

18 c)

If $T > 18$, then $T - 18 > 0$, so $\frac{dT}{dt} < 0$, meaning the tea is cooling down. The negative sign ensures the temperature decreases when the tea is hotter than the room.

18 d)

0.12 controls how quickly cooling happens. A larger value would mean faster cooling. Its units are per minute.

19 a)

$0.4P$ represents population growth proportional to the current population. In this context, as $0.4 > 0$, it would represent births.

19 b)

-2 represents a constant decrease in the population, such as a fixed number of fish being removed each year (e.g. fishing or harvesting).

19 c)

-2 does not depend on P as the model assumes the same number of fish are removed each year regardless of how many fish are in the lake, i.e. harvesting is assumed to be constant (not proportional).

20 a)

The term 6 is a constant positive value, so it represents a constant charge in coulombs per second.

20 b)

The term $\frac{Q}{4}$ increases as Q increases, so it represents an effect that becomes stronger the more charge is already stored. Since it is subtracted, it represents a factor that reduces the charging rate. So $\frac{Q}{4}$ represents a slowing effect that increases as the capacitor fills up.

20 c)

As Q increases, $\frac{Q}{4}$ increases, so the value of $6 - \frac{Q}{4}$ decreases. So the charging rate decreases as the capacitor stores more charge, which matches the realistic idea that it becomes harder to add more charge once it is already highly charged. So the negative term ensures the charging slows down as Q increases.

20 d)

As Q becomes large, $\frac{Q}{4}$ becomes large, so $6 - \frac{Q}{4}$ decreases. Eventually, when $6 - \frac{Q}{4} = 0 \Rightarrow Q = 24$. This means the capacitor approaches a maximum charge of 24 C and the charging rate $\frac{dQ}{dt}$ tends to zero.

21 a)

The equation is of the form $\frac{dS}{dt} = \text{rate in} - \text{rate out}$, so 0.8 is the rate at which salt enters the tank, i.e. 0.8 kg per minute.

21 b)

$\frac{5S}{100}$ must be the rate at which salt leaves the tank.

21 c)

In the $\frac{5S}{100}$ term, we can infer that because there is a constant denominator, the volume of liquid in the tank is remaining constant (at 100 litres). It is structurally of the form $5 \times \frac{S}{100}$, so liquid is exiting and entering the tank at 5 litres per minute.

21 d)

The inflow concentration is $\frac{0.8 \text{ kg per min}}{5 \text{ litres per min}} = 0.16$ kg per litre.

22 a)

The equation is of the form $\frac{dx}{dt} =$ rate in $-$ rate out, so 3.5 is the rate at which oxygen enters the tank, i.e. 3.5 cm^3 per hour.

22 b)

The equation is of the form $\frac{dx}{dt} =$ rate in $-$ rate out, so since the term $\frac{2x}{50+4t}$ is being subtracted, it represents the rate at which oxygen is leaving the tank.

22 c)

$$-0.5 = 3.5 - 2 \times \frac{150}{50+4t} \Rightarrow \frac{150}{50+4t} = 2 \Rightarrow \frac{75}{25+2t} = 2$$
$$75 = 50 + 4t \Rightarrow t = 6.25 \text{ hours} \Rightarrow t = 375 \text{ minutes.}$$

23 a)

$\frac{dS}{dt}$ means "rate of change of S with respect to time". Since S is measured in kg and t is measured in minutes, the units are kg per minute.

23 b)

Each term in the equation must have the same units as $\frac{dS}{dt}$, so 0.6 must also be measured in kg per minute, and represents the rate of salt entering the tank.

23 c)

Rewrite $\frac{3S}{40}$ as $3 \times \frac{S}{40}$. S is measured in kg. In a mixing problem, the 3 represents the outflow with units litres per minute. The 40 represents a constant volume, so 40 has units of litres. So $3 \times \frac{S}{40}$ has units (litres per minute) $\times \frac{\text{kg}}{\text{litres}} =$ kg per minute, which matches $\frac{dS}{dt}$.

24 a)

$\frac{dx}{dt}$ is the rate of change of mass with respect to time, so $\frac{dx}{dt}$ has units grams per minute. So the constant 1.5 must also have the units grams per minute.

24 b)

$\frac{4x}{80+2t}$ is the outflow term. x is measured in grams. $80 + 2t$ is measured in litres (representing the volume at time t). So $\frac{x}{80+2t}$ has units grams per litre. $\frac{4x}{80+2t}$ must have units grams per minute (to be consistent with $\frac{dx}{dt}$). So 4 must have units litres per minute (representing the outflow rate).

24 c)

If x were measured in kg, then $\frac{dx}{dt}$ would have units kg per minute. But the constant term 1.5 was interpreted as 1.5 grams per minute. So the left hand side would be kg per minute and the right hand side would be in grams per minute. This would make the equation dimensionally inconsistent. The constant term could be adjusted: 1.5 grams per minute = 0.0015 kg per minute.

25 a)

$\frac{dP}{dt}$ has units thousands of bacteria per hour.

25 b)

Each term in the differential equation must match the units of $\frac{dP}{dt}$, so k must also have units thousands of bacteria per hour.

25 c)

t is measured in hours, so $12+t$ must be measured in hours; therefore the number 12 must represent 12 hours.

25 d)

$$t = \frac{T}{60} \Rightarrow \frac{dt}{dT} = \frac{1}{60}, \quad \frac{dP}{dT} = \frac{dP}{dt} \times \frac{dt}{dT} = \frac{1}{60}\frac{dP}{dt}$$

$$\frac{dP}{dT} = \frac{1}{60}\left(k - \frac{P}{12+t}\right) = \frac{1}{60}\left(k - \frac{P}{12+\frac{T}{60}}\right) = \frac{1}{60}\left(k - \frac{60P}{720+T}\right)$$

$$= \frac{k}{60} - \frac{P}{720+T}.$$

26 a)

The volume decreases by 3 litres per minute, so $V = 90 - 3t$.

26 b)

$90 - 3t \geq 0 \Rightarrow t \leq 30$. So $0 \leq t \leq 30$ is the domain for which the model is physically valid.

26 c)

For $t > 30$, the formula gives $V < 0$, which is impossible as volume cannot be negative. The model is not valid for $t < 0$ as the situation is defined from $t = 0$.

27 a)

There are 6 litres flowing into the tank per minute, and 2 litres flowing out of the tank per minute, so the net inflow rate is 4 litres per minute: $V(t) = 40 + 6t - 2t \Rightarrow V(t) = 40 + 4t$.

27 b)

$120 = 40 + 4t \Rightarrow t = 20$. So the tank starts to overflow after 20 minutes.

27 c)

The model assumes the volume continues increasing, but this stops being true when the tank overflows. So the valid domain is $0 \leq t \leq 20$.

27 d)

The model assumes flow rates in and out of the tank are constant.

28 a)

$$\frac{dP}{dt} = 0.12P \Rightarrow \int \frac{1}{P}\,dP = \int 0.12\,dt \Rightarrow \ln|P| = 0.12t + c$$
$$\Rightarrow P = \pm e^{0.12t+c} \Rightarrow P = Ae^{0.12t}.$$
$$\text{When } t = 0,\ P = 500: \quad 500 = Ae^0 \Rightarrow A = 500, \quad \therefore P = 500e^{0.12t}.$$

28 b)

$P|_{t=20} = 500e^{0.12\times 20} = 5511.58819\ldots = 5510$ (3 SF).

28 c)

This is exponential growth, which assumes unlimited food, space and no disease. In reality, as the population grows, resources become limited, so the growth rate decreases.

28 d)

A better model would reduce the growth rate as the population becomes large, to reflect limited resources. One example would be $\frac{dP}{dt} = 0.12P - kP^2$.

29 a)

If $T > 20$ then $T - 20 > 0$ so $\frac{dT}{dt} < 0$, which implies the temperature decreases. If $T < 20$ then $T - 20 < 0$ so $\frac{dT}{dt} > 0$, which implies the temperature increases. So T approaches 20°C but does not go below it.

29 b)

$$\frac{dT}{dt} = -0.18(T - 20) \Rightarrow \int \frac{1}{T-20}\,dT = \int -0.18\,dt \Rightarrow \ln|T - 20| = -0.18t + c$$
$$\Rightarrow T - 20 = \pm e^{-0.18t+c} \Rightarrow M = 20 \pm e^{-0.18t+c} \Rightarrow M = 20 + Ae^{-0.18t}.$$
$$\text{When } t = 0,\ T = 70: \quad 70 = 20 + Ae^{-0.18\times 0} \Rightarrow A = 50, \quad \therefore M = 20 + 50e^{-0.18t}.$$

29 c)

Assumptions include: the room temperature stays constant at 20°C; the coffee is well-mixed so the temperature is uniform throughout the liquid; heat loss is proportional to the difference in the temperature.

30 a)

The equation contains $100 - t$ in the denominator, so we must have $t \neq 100$. As t is time, we must have $t \geq 0$, and hence $0 \leq t < 100$.

30 b)

The model is undefined at $t = 100$, since we have $100 - t$ in the denominator. Also, for $t > 100$, the denominator is negative, so $\frac{dM}{dt} > 0$, meaning the mass would increase rather than decay. Hence the model cannot be valid for all $t \geq 0$.

30 c)

$$\frac{dM}{dt} = -\frac{5M}{100-t} \Rightarrow \int \frac{1}{M}\,dM = \int -\frac{5}{100-t}\,dt$$
$$\Rightarrow \ln|M| = 5\ln|100-t| + c$$
$$\Rightarrow \ln|M| = \ln|(100-t)^5| + c$$
$$\Rightarrow M = \pm e^{\ln|(100-t)^5|+c}$$
$$\Rightarrow M = A(100-t)^5.$$

When $t = 0,\ M = 80$: $\quad 80 = A(100-0)^5$

$$\Rightarrow A = \frac{80}{100^5}$$
$$\therefore M = \frac{80(100-t)^5}{100^5}.$$

30 d)

The model is valid for $0 \leq t < 100$, so we consider t getting close to 100 but still less than 100. As $t \to 100$, $100 - t \to 0$, so $M = A(100-t)^5 \to 0$. Therefore the mass tends to zero.

30 e)

The model suggests that the chemical always reaches exactly zero mass at $t = 100$, which is unlikely in reality. Also, as $t \to 100$, the decay rate becomes infinite, which is physically impossible.

31 a)

$$\text{Net change in volume} = 4 - 4 = 0, \quad \therefore V = 120.$$
$$\text{Salt enters} = 4(0.25) = 1 \text{ kg/min}.$$
$$\text{Salt leaves} = 4\left(\frac{S}{120}\right) = \frac{S}{30} \text{ kg/min}.$$
$$\frac{dS}{dt} = \text{in} - \text{out} = 1 - \frac{S}{30}.$$

31 b)

$$\frac{dS}{dt} = 1 - \frac{S}{30}$$
$$\frac{dS}{dt} + \frac{S}{30} = 1$$

Integrating factor:

$$\text{IF} = e^{\int \frac{1}{30}\,dt} = e^{t/30}.$$
$$\frac{dS}{dt} + \frac{S}{30} = 1$$
$$e^{t/30}\frac{dS}{dt} + e^{t/30}\frac{S}{30} = e^{t/30}$$
$$\frac{d}{dt}(e^{t/30}S) = e^{t/30}$$

$$e^{t/30}S = 30e^{t/30} + c$$
$$S = 30 + ce^{-t/30}$$

Substitute $S = 10, t = 0$:

$$10 = 30 + ce^{-t/30}$$
$$10 = 30 + c$$
$$c = -20$$
$$S = 30 - 20e^{-t/30}.$$

31 c)

$S|_{t=20} = 30 - 20e^{-20/30} = 19.7$ kg (3 SF).

32 a)

$$\text{Net change in volume} = 8 - 6 = 2, \quad \therefore V = 50 + 2t.$$
$$\text{Salt enters} = 8(0.25) = 2 \text{ kg/min}.$$
$$\text{Salt leaves} = 6\left(\frac{S}{50+2t}\right) = \frac{6S}{50+2t} = \frac{3S}{25+t} \text{ kg/min}.$$
$$\frac{dS}{dt} = \text{in} - \text{out} = 2 - \frac{3S}{25+t}.$$

32 b)

$$\frac{dS}{dt} = 2 - \frac{3S}{25+t}$$
$$\frac{dS}{dt} + \frac{3S}{25+t} = 2$$

Integrating factor:

$$\text{IF} = e^{\int \frac{3}{25+t}\,dt} = e^{3\ln|25+t|} = e^{\ln|(25+t)^3|} = |(25+t)^3|.$$
$$\text{Let IF} = (25+t)^3.$$
$$\frac{dS}{dt} + \frac{3S}{25+t} = 2$$
$$(25+t)^3\frac{dS}{dt} + \frac{3S}{25+t}(25+t)^3 = 2(25+t)^3$$
$$(25+t)^3\frac{dS}{dt} + 3S(25+t)^2 = 2(25+t)^3$$
$$\frac{d}{dt}(S(25+t)^3) = 2(25+t)^3$$
$$S(25+t)^3 = \frac{1}{2}(25+t)^4 + c$$
$$S = \frac{1}{2}(25+t) + c(25+t)^{-3}$$

Substitute $S = 5.5, t = 0$:

$$5.5 = \frac{1}{2}(25+0) + c(25+0)^{-3}$$

$$
\begin{aligned}
5.5 &= \frac{25}{2} + \frac{c}{15625} \\
c &= -109375 \\
S &= \frac{1}{2}(25+t) - 109375(25+t)^{-3} \\
&= \frac{1}{2}(25+t) - \frac{109375}{(25+t)^3}.
\end{aligned}
$$

32 c)

The incoming concentration is 0.25 kg per litre. Over time, the tank becomes dominated by the inflow mixture, so concentration tends to 0.25 kg per litre. Alternatively, the concentration is

$$
\begin{aligned}
\frac{S}{V} &= \frac{S}{50+2t} = \frac{S}{2(25+t)} \\
&= \frac{\frac{1}{2}(25+t) - \frac{109375}{(25+t)^3}}{2(25+t)} \\
&= \frac{1}{4} - \frac{109375}{2(25+t)^4}.
\end{aligned}
$$

As $t \to \infty$, $\frac{109375}{2(25+t)^4} \to 0$, so concentration tends to $\frac{1}{4} = 0.25$ kg per litre.

32 d)

The model assumes that the salt is perfectly mixed with the water, so the concentration is uniform throughout the tank at all times.

33 a)

$$
\begin{aligned}
&\text{Net change in volume} = 70 - 30 = 40, \quad V = 600 + 40t. \\
&\text{Carbon dioxide enters} = 70(0.08) = 5.6 \text{ cm}^3\text{/hour}. \\
&\text{Carbon dioxide leaves} = 30\left(\frac{x}{600+40t}\right) = \frac{30x}{600+40t} = \frac{3x}{60+4t} \text{ cm}^3\text{/hour}. \\
&\frac{dx}{dt} = \text{in} - \text{out} = 5.6 - \frac{3x}{60+4t}.
\end{aligned}
$$

33 b)

$$
\begin{aligned}
\frac{dx}{dt} + \frac{3x}{60+4t} &= 5.6 \\
\text{IF} &= e^{\int \frac{3}{60+4t}\,dt} = e^{\frac{3}{4}\ln|60+4t|} = |(60+4t)^{3/4}| \\
\text{Let IF} &= (60+4t)^{3/4}. \\
(60+4t)^{3/4}\frac{dx}{dt} + \frac{3x}{60+4t}(60+4t)^{3/4} &= 5.6(60+4t)^{3/4} \\
(60+4t)^{3/4}\frac{dx}{dt} + 3x(60+4t)^{-1/4} &= 5.6(60+4t)^{3/4} \\
\frac{d}{dt}\left(x(60+4t)^{3/4}\right) &= 5.6(60+4t)^{3/4} \\
x(60+4t)^{3/4} &= 0.8(60+4t)^{7/4} + c
\end{aligned}
$$

$$x = 0.8(60 + 4t) + c(60 + 4t)^{-3/4}$$

Initially the vessel contains only nitrogen, so $x = 0$ at $t = 0$:

$$\begin{aligned}
0 &= 0.8(60) + c(60)^{-3/4} \\
0 &= 48 + 60^{-3/4}c \\
c &= -48(60)^{3/4} \\
x &= 0.8(60 + 4t) - 48(60)^{3/4}(60 + 4t)^{-3/4} \\
&= 0.8(60 + 4t) - 48\left(\frac{60}{60 + 4t}\right)^{3/4}.
\end{aligned}$$

At $t = 5$:

$$\begin{aligned}
x|_{t=5} &= 0.8(60 + 4 \times 5) - 48\left(\frac{60}{60 + 4 \times 5}\right)^{3/4} \\
&= 64 - 38.684\ldots \\
&= 25.3 \text{ cm}^3 \text{ (3 SF)}.
\end{aligned}$$

33 c)

The incoming concentration of salt in the water is 0.25 kg per litre. Over time, the tank becomes dominated by the inflow mixture, so the concentration tends towards 0.25 kg per litre. Alternatively, the concentration is

$$\begin{aligned}
\frac{x}{V} = \frac{x}{600 + 40t} &= \frac{x}{10(60 + 4t)} \\
&= \frac{0.8(60 + 4t) - 48\left(\frac{60}{60+4t}\right)^{3/4}}{10(60 + 4t)} \\
&= 0.08 - \frac{48}{10}\frac{60^{3/4}}{(60 + 4t)^{7/4}}.
\end{aligned}$$

As $t \to \infty$, the second term tends to zero, so the concentration tends to $0.08 = 8\%$.

33 d)

Possible limitations include: the gases may not mix instantly and uniformly throughout the vessel; the flow and leak rates may not remain constant as pressure changes; the vessel cannot increase in volume indefinitely; the incoming percentage of carbon dioxide may not remain constant.

34 a)

$$\begin{aligned}
\text{Net change in volume} &= 3 + 1 - 4 = 0. \\
\text{Volume of ethanol and water} &= 15 + 20 = 40 \text{ litres}. \\
\text{Fraction ethanol in vessel} &= \frac{x}{40}. \\
\text{Ethanol leaves} &= 4\left(\frac{x}{40}\right) = \frac{x}{10} \text{ litres/s}. \\
\frac{dx}{dt} &= \text{in} - \text{out} = 3 - \frac{x}{10}, \quad \text{so } k = 10.
\end{aligned}$$

34 b)

$$\begin{aligned}
\frac{dx}{dt} + \frac{x}{10} &= 3 \\
\text{IF} &= e^{\int \frac{1}{10}\,dt} = e^{t/10} \\
e^{t/10}\frac{dx}{dt} + \frac{x}{10}e^{t/10} &= 3e^{t/10} \\
\frac{d}{dt}(xe^{t/10}) &= 3e^{t/10} \\
xe^{t/10} &= 30e^{t/10} + c \\
x &= 30 + ce^{-t/10}
\end{aligned}$$

Initially $x = 15$ litres at $t = 0$:

$$\begin{aligned}
15 &= 30 + ce^0 \\
c &= -15 \\
x &= 30 - 15e^{-t/10}.
\end{aligned}$$

For the ethanol-to-water ratio to be $2:1$, ethanol volume is $\frac{2}{3}(40) = \frac{80}{3}$ litres. Substitute $x = \frac{80}{3}$:

$$\begin{aligned}
\frac{80}{3} &= 30 - 15e^{-t/10} \\
e^{-t/10} &= \frac{2}{9} \\
-\frac{t}{10} &= \ln\frac{2}{9} \\
t &= -10\ln\frac{2}{9} \\
&= 15 \text{ seconds to the nearest second.}
\end{aligned}$$

34 c)

The model predicts the ratio $2:1$ occurs after about 15 seconds, while experimentally it occurs after 18 seconds. These differ, so the model is not fully valid. A possible reason is that the mixture may not be perfectly uniform at all times, even though it is stirred.

35 a)

$$\text{Net rate of change of volume} = 0.1 - 0.05 = 0.05 \text{ litres/min.}$$
$$V(t) = 5 + 0.05t.$$

35 b)

The inflow concentration is $2 + 3e^{-0.08t}$ mg/litre and inflow rate is 0.1 litres/min, so rate in is $0.1(2 + 3e^{-0.08t})$ mg/min. The concentration inside the tank is $m/V(t)$ mg/litre. Outflow is 0.05 litres/min, so

$$\text{rate out} = 0.05\frac{m}{V(t)} = 0.05\frac{m}{5 + 0.05t} = \frac{1}{20}\frac{m}{5 + t/20} = \frac{m}{100 + t}.$$
$$\frac{dm}{dt} = \text{rate in} - \text{rate out}$$

$$= 0.1(2 + 3e^{-0.08t}) - \frac{m}{100+t}$$
$$\frac{dm}{dt} + \frac{m}{100+t} = \frac{1}{10}(2 + 3e^{-0.08t}).$$

35 c)

$$\text{IF} = e^{\int \frac{1}{100+t}\,dt} = e^{\ln|100+t|} = |100+t|.$$
$$\text{Let IF} = 100 + t.$$
$$\frac{dm}{dt} + \frac{m}{100+t} = \frac{1}{10}(2 + 3e^{-0.08t})$$
$$(100+t)\frac{dm}{dt} + m = \frac{1}{10}(100+t)(2 + 3e^{-0.08t})$$
$$\frac{d}{dt}((100+t)m) = \frac{1}{10}(100+t)(2 + 3e^{-0.08t})$$

Let $I = \int (100+x)(2+3e^{-0.08x})\,dx$. By parts, $u = 100+x,\ dv/dx = 2+3e^{-0.08x},\ du/dx = 1,\ v = 2x - \frac{75}{2}e^{-0.08x}$:

$$I = (100+x)(2x - \tfrac{75}{2}e^{-0.08x}) - \int (2x - \tfrac{75}{2}e^{-0.08x})\,dx$$
$$= (100+x)(2x - \tfrac{75}{2}e^{-0.08x}) - (x^2 + \tfrac{1875}{4}e^{-0.08x}) + k.$$

Therefore

$$(100+t)m = \frac{1}{10}\left((100+t)(2t - \frac{75}{2}e^{-0.08t}) - (t^2 + \frac{1875}{4}e^{-0.08t})\right) + c$$
$$m = \frac{1}{10}(2t - \frac{75}{2}e^{-0.08t}) + \frac{c - (t^2 + \frac{1875}{4}e^{-0.08t})}{1000 + 10t}.$$

Initially the vessel contains 5 litres of pure water, so $m = 0, t = 0$:

$$0 = \frac{1}{10}(0 - \frac{75}{2}) + \frac{c - \frac{1875}{4}}{1000}$$
$$0 = -\frac{15}{4} + \frac{c - \frac{1875}{4}}{1000}$$
$$c - \frac{1875}{4} = 3750, \quad c = \frac{16875}{4}.$$

Thus

$$m = \frac{1}{10}(2t - \frac{75}{2}e^{-0.08t}) + \frac{\frac{16875}{4} - (t^2 + \frac{1875}{4}e^{-0.08t})}{1000 + 10t}$$
$$= \frac{1}{5}t - \frac{15}{4}e^{-0.08t} + \frac{16875 - 4t^2 - 1875e^{-0.08t}}{4000 + 40t}.$$

35 d)

At $t = 25$,

$$V(25) = 5 + 0.05(25) = 6.25 \text{ litres},$$
$$m(25) = \frac{1}{5}(25) - \frac{15}{4}e^{-0.08(25)} + \frac{16875 - 4(25)^2 - 1875e^{-0.08(25)}}{4000 + 40(25)}$$

$$= 5 - 0.5075073121\ldots + \frac{14121.24634\ldots}{5000}$$
$$= 7.316741957\ldots \text{ mg.}$$

Therefore concentration $= m(25)/V(25) = 7.316741957\ldots/6.25 = 1.17$ mg/litre. The actual figure was 1.52 mg/litre, a difference $(1.52 - 1.17)/1.52 \times 100 \approx 23\%$. The model underestimates concentration; one possible reason is that the pure water and chemical solution are not perfectly mixed.

36)

$$\frac{dT}{dt} + T = 4e^t \cos t, \quad \text{IF} = e^{\int 1\,dt} = e^t.$$
$$e^t \frac{dT}{dt} + Te^t = 4e^{2t} \cos t$$
$$\frac{d}{dt}(Te^t) = 4e^{2t} \cos t.$$

Let $I = \int e^{2x} \cos x\,dx$. By parts, $u = \cos x,\ dv/dx = e^{2x},\ du/dx = -\sin x,\ v = \frac{1}{2}e^{2x}$:

$$I = \frac{1}{2}e^{2x} \cos x + \frac{1}{2}\int e^{2x} \sin x\,dx.$$

Use integration by parts again, $u = \sin x,\ dv/dx = \frac{1}{2}e^{2x},\ du/dx = \cos x,\ v = \frac{1}{4}e^{2x}$:

$$I = \frac{1}{2}e^{2x} \cos x + \frac{1}{4}e^{2x} \sin x - \int \frac{1}{4}e^{2x} \cos x\,dx$$
$$= \frac{1}{2}e^{2x} \cos x + \frac{1}{4}e^{2x} \sin x - \frac{1}{4}I$$
$$\frac{5}{4}I = \frac{1}{2}e^{2x} \cos x + \frac{1}{4}e^{2x} \sin x + k_1$$
$$I = \frac{2}{5}e^{2x} \cos x + \frac{1}{5}e^{2x} \sin x + k_2.$$

Hence

$$Te^t = 4\left(\frac{2}{5}e^{2t} \cos t + \frac{1}{5}e^{2t} \sin t\right) + c$$
$$T = \frac{8}{5}e^t \cos t + \frac{4}{5}e^t \sin t + ce^{-t}.$$

When $t = 0, T = 0$:

$$0 = \frac{8}{5} + \frac{4}{5}(0) + c$$
$$c = -\frac{8}{5}$$
$$T = \frac{8}{5}e^t \cos t + \frac{4}{5}e^t \sin t - \frac{8}{5}e^{-t}.$$

Thus

$$T|_{t=\pi} = -\frac{8}{5}(e^\pi + e^{-\pi})^\circ\text{C}.$$

37 a)

By separation of variables,

$$\begin{aligned}\frac{dN}{dt} &= -kN \\ \int \frac{1}{N}\,dN &= \int -k\,dt \\ \ln|N| &= -kt + c \\ N &= Ae^{-kt}.\end{aligned}$$

At $t = 0, N = N_0$, so $A = N_0$ and $N = N_0e^{-kt}$. At $t = 1, N = 0.88N_0$:

$$\begin{aligned}0.88N_0 &= N_0e^{-k} \\ 0.88 &= e^{-k} \\ k &= -\ln 0.88 = 0.128 \text{ (3 SF)}.\end{aligned}$$

37 b)

Substituting $k = 0.128$ into the model gives $\frac{dN}{dt} = -0.128N + 30t + 60$.

37 c)

$$\begin{aligned}\frac{dN}{dt} + 0.128N &= 30t + 60 \\ \text{IF} &= e^{\int 0.128\,dt} = e^{0.128t} \\ e^{0.128t}\frac{dN}{dt} + 0.128e^{0.128t}N &= 30te^{0.128t} + 60e^{0.128t} \\ \frac{d}{dt}(e^{0.128t}N) &= 30te^{0.128t} + 60e^{0.128t}.\end{aligned}$$

Let $I = \int 30xe^{0.128x}\,dx$. By parts, $u = 30x,\ dv/dx = e^{0.128x},\ du/dx = 30,\ v = e^{0.128x}/0.128$:

$$\begin{aligned}I &= \frac{1875xe^{0.128x}}{8} - \int \frac{1875e^{0.128x}}{8}\,dx \\ &= \frac{1875xe^{0.128x}}{8} - \frac{1875}{8}\frac{e^{0.128x}}{0.128} + k \\ &= \frac{1875xe^{0.128x}}{8} - \frac{234375e^{0.128x}}{128} + k.\end{aligned}$$

Therefore

$$\begin{aligned}e^{0.128t}N &= \frac{1875te^{0.128t}}{8} - \frac{234375e^{0.128t}}{128} + \frac{1875}{4}e^{0.128t} + c \\ &= \frac{1875te^{0.128t}}{8} - \frac{174375e^{0.128t}}{128} + c \\ N &= \frac{1875t}{8} - \frac{174375}{128} + ce^{-0.128t}.\end{aligned}$$

At $t = 0, N = 500$:

$$\begin{aligned}500 &= 0 - \frac{174375}{128} + c \\ c &= \frac{238375}{128} \\ N &= \frac{1875t}{8} - \frac{174375}{128} + \frac{238375}{128}e^{-0.128t} \\ &= 234.375t - 1362.30 + 1862.30e^{-0.128t}.\end{aligned}$$

37 d)

Possible limitations include:

- The model assumes the leaving rate is always proportional to N, but in reality customers may leave because of competition, pricing, or customer satisfaction.
- The model assumes that the marketing term $30t$ increases forever, which is unrealistic as the market is finite.
- The model assumes the partnerships add exactly 60 subscribers per year continuously, which may fluctuate.
- The model ignores seasonal effects, and assumes subscribers join and leave continuously, rather than in discrete monthly renewals.

38 a)

$$\left(\frac{4+2t}{\sqrt{t}}\right)\frac{dM}{dt}+\frac{4M}{\sqrt{t}}=105(2+t)$$
$$(4+2t)\frac{dM}{dt}+4M=105(2+t)\sqrt{t}$$
$$\frac{dM}{dt}+\frac{4M}{4+2t}=\frac{105(2+t)\sqrt{t}}{4+2t}$$
$$\frac{dM}{dt}+\frac{2M}{2+t}=\frac{105}{2}\sqrt{t}$$

Integrating factor:

$$\text{IF}=e^{\int\frac{2}{2+t}\,dt}=e^{2\ln|2+t|}=e^{\ln|(2+t)^2|}=|(2+t)^2|.$$
$$\text{Let IF}=(2+t)^2.$$
$$\frac{dM}{dt}+\frac{2M}{2+t}=\frac{105}{2}\sqrt{t}$$
$$(2+t)^2\frac{dM}{dt}+\frac{2M}{2+t}(2+t)^2=\frac{105}{2}(2+t)^2\sqrt{t}$$
$$(2+t)^2\frac{dM}{dt}+2(2+t)M=\frac{105}{2}(2+t)^2\sqrt{t}$$
$$\frac{d}{dt}((2+t)^2M)=\frac{105}{2}(2+t)^2\sqrt{t}$$
$$=\frac{105}{2}(4+4t+t^2)t^{1/2}$$
$$=210t^{1/2}+210t^{3/2}+\frac{105}{2}t^{5/2}$$
$$(2+t)^2M=140t^{3/2}+84t^{5/2}+15t^{7/2}+c$$
$$M=\frac{140t^{3/2}+84t^{5/2}+15t^{7/2}+c}{(2+t)^2}.$$

When $t=0, M=4$:

$$4=\frac{c}{(2+0)^2}$$
$$c=16$$
$$M=\frac{140t^{3/2}+84t^{5/2}+15t^{7/2}+16}{(2+t)^2}.$$

When $t=6$:

$$M|_{t=6}=\frac{140(6)^{3/2}+84(6)^{5/2}+15(6)^{7/2}+16}{(2+6)^2}$$

$$= \frac{17417.17513\ldots}{64}$$
$$= 272 \text{ grams (3 SF)}.$$

38 b)

$$\left(\frac{4+2t}{\sqrt{t}}\right)\frac{dM}{dt} + \frac{4M}{\sqrt{t}} = 105(2+t)$$
$$\left(\frac{4+2t}{\sqrt{t}}\right)\frac{dM}{dt} = 105(2+t) - \frac{4M}{\sqrt{t}}$$
$$\frac{dM}{dt} = \left(105(2+t) - \frac{4M}{\sqrt{t}}\right)\left(\frac{\sqrt{t}}{4+2t}\right).$$

At $t = 3$:

$$M|_{t=3} = \frac{140(3)^{3/2} + 84(3)^{5/2} + 15(3)^{7/2} + 16}{(2+3)^2}$$
$$= \frac{2754.372327\ldots}{25}$$
$$= 110.1748931\ldots \text{ grams},$$
$$\left.\frac{dM}{dt}\right|_{t=3} = \left(105(2+3) - \frac{4(110.1748931\ldots)}{\sqrt{3}}\right)\left(\frac{\sqrt{3}}{4+2(3)}\right)$$
$$= 46.9 \text{ grams per hour (3 SF)}.$$

38 c)

The model assumes that the impurity is evenly mixed throughout the system at all times, which may not be true in practice.

39)

We are told the rate of decrease is proportional to $x^2 + 16$, so

$$\frac{dx}{dt} = -k(x^2+16), \quad k > 0.$$

Separate variables:

$$\int \frac{1}{x^2+16}\,dx = \int -k\,dt$$
$$\int \frac{1}{16(1+\frac{1}{16}x^2)}\,dx = \int -k\,dt$$
$$\frac{1}{16}\int \frac{1}{1+(\frac{1}{4}x)^2}\,dx = \int -k\,dt$$
$$\frac{1}{4}\arctan\left(\frac{1}{4}x\right) = -kt + c.$$

Substitute $x = 120, t = 0$:

$$\frac{1}{4}\arctan(30) = c.$$

Substitute $x = 60, t = 50$:

$$\begin{aligned}\frac{1}{4}\arctan(15) &= -50k + \frac{1}{4}\arctan(30)\\ k &= \frac{1}{200}\arctan(30) - \frac{1}{200}\arctan(15)\\ &= \frac{1}{200}(\arctan(30) - \arctan(15)).\end{aligned}$$

Therefore

$$\begin{aligned}\frac{1}{4}\arctan\left(\frac{1}{4}x\right) &= -\frac{1}{200}(\arctan(30) - \arctan(15))t + \frac{1}{4}\arctan(30)\\ \arctan\left(\frac{1}{4}x\right) &= \frac{1}{50}(\arctan(15) - \arctan(30))t + \arctan(30)\\ x &= 4\tan\left(\frac{1}{50}(\arctan(15) - \arctan(30))t + \arctan(30)\right).\end{aligned}$$

At $t = 200$,

$$x|_{t=200} = 4\tan\left(\frac{1}{50}(\arctan(15) - \arctan(30))(200) + \arctan(30)\right) = 23.8 \text{ mg (3 SF)}.$$

40 a)

$$\begin{aligned}&\text{Driving force} = (6 + 2t)\text{ N}, \quad \text{resistive force} = 3v\text{ N},\\ &\text{Mass} = 3\text{ kg}, \quad a = \frac{dv}{dt},\\ F = ma: \quad &(6 + 2t) - 3v = 3\frac{dv}{dt}\\ &\frac{dv}{dt} = 2 + \frac{2}{3}t - v\\ &\frac{dv}{dt} + v = 2 + \frac{2}{3}t.\end{aligned}$$

40 b)

$$\begin{aligned}\frac{dv}{dt} + v &= 2 + \frac{2}{3}t\\ \text{IF} &= e^{\int 1\,dt} = e^t\\ e^t\frac{dv}{dt} + e^t v &= 2e^t + \frac{2}{3}te^t\\ \frac{d}{dt}(e^t v) &= 2e^t + \frac{2}{3}te^t.\end{aligned}$$

Let $I = \int \frac{2}{3}xe^x\,dx$. By parts, $u = \frac{2}{3}x,\ dv/dx = e^x,\ du/dx = \frac{2}{3},\ v = e^x$:

$$\begin{aligned}I &= \frac{2}{3}xe^x - \int \frac{2}{3}e^x\,dx\\ &= \frac{2}{3}xe^x - \frac{2}{3}e^x + k.\end{aligned}$$

Therefore

$$\begin{aligned} e^t v &= 2e^t + \frac{2}{3}te^t - \frac{2}{3}e^t + c \\ &= \frac{4}{3}e^t + \frac{2}{3}te^t + c \\ v &= \frac{4}{3} + \frac{2}{3}t + ce^{-t}. \end{aligned}$$

When $t = 0, v = 0$:

$$\begin{aligned} 0 &= \frac{4}{3} + 0 + c \\ c &= -\frac{4}{3} \\ v &= \frac{4}{3} + \frac{2}{3}t - \frac{4}{3}e^{-t}. \end{aligned}$$

40 c)

When $t = 3$,

$$v|_{t=3} = \frac{4}{3} + \frac{2}{3}(3) - \frac{4}{3}e^{-3} = \left(\frac{10}{3} - \frac{4}{3}e^{-3}\right) \text{ m s}^{-1}.$$

40 d)

$$\begin{aligned} &\text{Driving force} = 9 \text{ N}, \quad \text{resistive force} = 3v \text{ N}, \\ &\text{Mass} = 3 \text{ kg}, \quad a = \frac{dv}{dt}, \\ F = ma: \quad &9 - 3v = 3\frac{dv}{dt} \\ &\frac{dv}{dt} = 3 - v. \end{aligned}$$

40 e)

Method 1: Use an integrating factor

$$\begin{aligned} \frac{dv}{dt} + v = 3, \quad &\text{IF} = e^t \\ e^t\frac{dv}{dt} + e^t v &= 3e^t \\ \frac{d}{dt}(e^t v) &= 3e^t \\ e^t v &= 3e^t + c \\ v &= 3 + ce^{-t}. \end{aligned}$$

When $t = 3, v = \frac{10}{3} - \frac{4}{3}e^{-3}$:

$$\begin{aligned} \frac{10}{3} - \frac{4}{3}e^{-3} &= 3 + ce^{-3} \\ \frac{10}{3}e^3 - \frac{4}{3} &= 3e^3 + c \\ c &= \frac{1}{3}e^3 - \frac{4}{3} \end{aligned}$$

$$v = 3 + \left(\frac{1}{3}e^3 - \frac{4}{3}\right)e^{-t}.$$

Method 2: Separation of variables

$$\begin{aligned}
\int \frac{1}{3-v}\,dv &= \int 1\,dt \\
-\ln|3-v| &= t + c \\
\ln|3-v| &= k - t \\
3 - v &= \pm e^{k-t} \\
3 - v &= Ae^{-t} \\
v &= 3 - Ae^{-t}.
\end{aligned}$$

When $t = 3, v = \frac{10}{3} - \frac{4}{3}e^{-3}$:

$$\begin{aligned}
\frac{10}{3} - \frac{4}{3}e^{-3} &= 3 - Ae^{-3} \\
\frac{10}{3}e^3 - \frac{4}{3} &= 3e^3 - A \\
A &= \frac{1}{3}e^3 - \frac{4}{3} \\
v &= 3 - \left(\frac{1}{3}e^3 - \frac{4}{3}\right)e^{-t}.
\end{aligned}$$

41 a)

$$\text{Driving force} = P = kt \text{ N}, \quad \text{resistive force} = Q = cu \text{ N},$$
$$\text{Mass} = 0.4 \text{ kg}, \quad \text{acceleration } a = \frac{du}{dt}.$$
$$\begin{aligned}
F = ma: \quad kt - cu &= 0.4\frac{du}{dt} \\
\frac{2}{5}\frac{du}{dt} &= kt - cu \\
\frac{du}{dt} &= \frac{5}{2}(kt - cu).
\end{aligned}$$

41 b)

$$\begin{aligned}
\frac{du}{dt} &= \frac{5}{2}(kt - cu) \\
\frac{du}{dt} + \frac{5}{2}cu &= \frac{5}{2}kt \\
\text{IF} &= e^{\int \frac{5}{2}c\,dt} = e^{5ct/2} \\
e^{5ct/2}\frac{du}{dt} + \frac{5}{2}ce^{5ct/2}u &= \frac{5}{2}kte^{5ct/2} \\
\frac{d}{dt}(e^{5ct/2}u) &= \frac{5}{2}kte^{5ct/2}.
\end{aligned}$$

Let $I = \int \frac{5}{2}kxe^{5cx/2}\,dx$. By parts, $u = \frac{5}{2}kx,\ dv/dx = e^{5cx/2},\ du/dx = \frac{5}{2}k,\ v = \frac{2}{5c}e^{5cx/2}$:

$$I = \frac{k}{c}xe^{5cx/2} - \int \frac{k}{c}e^{5cx/2}\,dx$$

$$= \frac{k}{c}xe^{5cx/2} - \frac{2k}{5c^2}e^{5cx/2} + c_1.$$

Therefore

$$e^{5ct/2}u = \frac{k}{c}te^{5ct/2} - \frac{2k}{5c^2}e^{5ct/2} + c_2$$
$$u = \frac{k}{c}t - \frac{2k}{5c^2} + c_2e^{-5ct/2}.$$

When $t = 0, u = 0$:

$$0 = 0 - \frac{2k}{5c^2} + c_2$$
$$c_2 = \frac{2k}{5c^2}$$
$$u = \frac{k}{c}t - \frac{2k}{5c^2} + \frac{2k}{5c^2}e^{-5ct/2}.$$

41 c)

Substitute $k = 1.6, c = 0.8$:

$$u = \frac{1.6}{0.8}t - \frac{2(1.6)}{5(0.8)^2} + \frac{2(1.6)}{5(0.8)^2}e^{-5(0.8)t/2}$$
$$= 2t - 1 + e^{-2t}.$$

Acceleration:

$$a = \frac{du}{dt} = 2 - 2e^{-2t}$$
$$a|_{t=2} = 2 - 2e^{-2(2)}$$
$$= 1.96 \text{ m s}^{-2} \text{ (3 SF)}.$$

41 d)

$$\text{Driving force} = P = 1.6 \text{ N}, \quad \text{resistive force} = Q = 0.8u \text{ N},$$
$$\text{Mass} = 0.4 \text{ kg}, \quad \text{acceleration } a = \frac{du}{dt}.$$
$$F = ma: \quad 1.6 - 0.8u = 0.4\frac{du}{dt}$$
$$\frac{2}{5}\frac{du}{dt} = 1.6 - 0.8u$$
$$\frac{du}{dt} = 4 - 2u.$$

41 e)

The original model was

$$\frac{du}{dt} = 4t - 2u \quad \Rightarrow \quad u = 2t - 1 + e^{-2t}.$$

The refined model is

$$\frac{du}{dt} = 4 - 2u \quad \Rightarrow \quad u = 2 - 2e^{-2t}.$$

In the original model, the driving force continues to increase over time without bound. In the refined model, the diving force is constant, so speed tends to a finite limiting value. The speed predicted by the initial model will eventually exceed that of the refined model.

42 a)

Downwards is positive. The downward force is the weight mg N and the upward resistance is $km(v+2)$ N. With mass m and acceleration dv/dt:

$$\begin{aligned} mg - km(v+2) &= m\frac{dv}{dt} \\ \frac{dv}{dt} &= g - k(v+2) \\ &= g - kv - 2k \\ \frac{dv}{dt} + kv &= g - 2k. \end{aligned}$$

42 b)

$$\begin{aligned} \frac{dv}{dt} + kv &= g - 2k \\ \text{IF} &= e^{\int k\,dt} = e^{kt} \\ e^{kt}\frac{dv}{dt} + ke^{kt}v &= (g-2k)e^{kt} \\ \frac{d}{dt}(e^{kt}v) &= (g-2k)e^{kt} \\ e^{kt}v &= \frac{g-2k}{k}e^{kt} + c \\ v &= \frac{g-2k}{k} + ce^{-kt}. \end{aligned}$$

When $t = 0, v = u$:

$$\begin{aligned} u &= \frac{g-2k}{k} + c \\ c &= u - \frac{g-2k}{k} \\ v &= \frac{g-2k}{k} + \left(u - \frac{g-2k}{k}\right)e^{-kt}. \end{aligned}$$

42 c)

As $t \to \infty, e^{-kt} \to 0$, so $v \to \frac{g-2k}{k}$ m s^{-1}. Thus terminal velocity is $v_T = \frac{g-2k}{k}$ m s^{-1}, provided $g > 2k$; otherwise the model suggests a negative terminal velocity, inconsistent with downward motion.

42 d)

Downwards is positive. The downward force is mg N; resistance upwards is $km\sqrt{v}$ N. With mass m kg and acceleration dv/dt,

$$\begin{aligned} mg - km\sqrt{v} &= m\frac{dv}{dt} \\ \frac{dv}{dt} &= g - k\sqrt{v}. \end{aligned}$$

42 e)

The limiting velocity occurs when acceleration tends to zero:

$$\begin{aligned} \frac{dv}{dt} &= 0 \\ g - k\sqrt{v} &= 0 \\ k\sqrt{v} &= g \\ \sqrt{v} &= \frac{g}{k} \\ v &= \frac{g^2}{k^2}. \end{aligned}$$

Consider $v < g^2/k^2$:

$$\sqrt{v} < g/k \Rightarrow g - k\sqrt{v} > 0 \Rightarrow dv/dt > 0,$$

so velocity increases below g^2/k^2. For $v > g^2/k^2$:

$$\sqrt{v} > g/k \Rightarrow g - k\sqrt{v} < 0 \Rightarrow dv/dt < 0,$$

so velocity decreases above g^2/k^2. Therefore the velocity tends to g^2/k^2 m s^{-1} as $t \to \infty$.

43 a)

The carriage starts from rest at $t = 0$, so $v = 0$. Therefore the resistive force $0.12m(0) = 0$ N.

43 b)

For $6 \leq t \leq 10$:

$$\begin{aligned} &\text{Engine force} = 0 \text{ N}, \quad \text{buffer force} = 1.8m \text{ N}, \\ &\text{resistance} = 0.12mv \text{ N}, \quad \text{mass} = m, \quad a = \frac{dv}{dt}. \\ F = ma: \quad -1.8m - 0.12mv &= m\frac{dv}{dt} \\ \frac{dv}{dt} &= -1.8 - 0.12v \\ \frac{dv}{dt} + 0.12v &= -1.8. \end{aligned}$$

43 c)

$$\begin{aligned} \frac{dv}{dt} + 0.12v &= -1.8 \\ \text{IF} &= e^{\int 0.12\,dt} = e^{0.12t} \\ e^{0.12t}\frac{dv}{dt} + 0.12e^{0.12t}v &= -1.8e^{0.12t} \\ \frac{d}{dt}(e^{0.12t}v) &= -1.8e^{0.12t} \\ e^{0.12t}v &= -15e^{0.12t} + k_1 \\ v &= -15 + k_1e^{-0.12t}. \end{aligned}$$

When $t = 10, v = 0$:

$$0 = -15 + k_1e^{-0.12(10)}$$

$$\begin{aligned} k_1 &= 15e^{1.2} \\ v &= -15 + 15e^{1.2}e^{-0.12t} \\ &= -15 + 15e^{1.2-0.12t}. \end{aligned}$$

43 d)

When $t = 6$,

$$\begin{aligned} v|_{t=6} &= -15 + 15e^{1.2-0.12(6)} \\ &= -15 + 15e^{0.48} \\ &= 9.24 \text{ m s}^{-1} \text{ (3 SF)}. \end{aligned}$$

43 e)

For $0 \le t \le 6$:

$$\begin{aligned} &\text{Engine force} = ct \text{ N}, \quad \text{resistance} = 0.12mv \text{ N}, \\ &\text{mass} = m, \quad a = \frac{dv}{dt}. \\ F = ma: \quad ct - 0.12mv &= m\frac{dv}{dt} \\ \frac{dv}{dt} &= \frac{c}{m}t - 0.12v \\ \frac{dv}{dt} + 0.12v &= \frac{c}{m}t. \end{aligned}$$

43 f)

$$\begin{aligned} \frac{dv}{dt} + 0.12v &= \frac{c}{m}t \\ \text{IF} &= e^{\int 0.12\,dt} = e^{0.12t} \\ e^{0.12t}\frac{dv}{dt} + 0.12e^{0.12t}v &= \frac{c}{m}te^{0.12t} \\ \frac{d}{dt}(e^{0.12t}v) &= \frac{c}{m}te^{0.12t}. \end{aligned}$$

Let $I = \int \frac{c}{m}xe^{0.12x}\,dx$. By parts, $u = \frac{c}{m}x$, $dv/dx = e^{0.12x}$, $du/dx = \frac{c}{m}$, $v = \frac{25}{3}e^{0.12x}$:

$$\begin{aligned} I &= \frac{25c}{3m}xe^{0.12x} - \int \frac{25c}{3m}e^{0.12x}\,dx \\ &= \frac{25c}{3m}xe^{0.12x} - \frac{625c}{9m}e^{0.12x} + k_2. \end{aligned}$$

Therefore

$$\begin{aligned} e^{0.12t}v &= \frac{25c}{3m}te^{0.12t} - \frac{625c}{9m}e^{0.12t} + k_3 \\ v &= \frac{25c}{3m}t - \frac{625c}{9m} + k_3e^{-0.12t}. \end{aligned}$$

When $t = 0, v = 0$:

$$0 = 0 - \frac{625c}{9m} + k_3$$

$$k_3 = \frac{625c}{9m}$$
$$v = \frac{25c}{3m}t - \frac{625c}{9m} + \frac{625c}{9m}e^{-0.12t}.$$

43 g)

At $t = 6$, the velocities must be equal. From earlier,

$$v|_{t=6} = -15 + 15e^{0.48}.$$

Also

$$\begin{aligned} v|_{t=6} &= \frac{25c}{3m}(6) - \frac{625c}{9m} + \frac{625c}{9m}e^{-0.12(6)} \\ &= \frac{50c}{m} - \frac{625c}{9m} + \frac{625c}{9m}e^{-0.72} \\ &= -\frac{175c}{9m} + \frac{625c}{9m}e^{-0.72} \\ &= \frac{c}{m}\left(-\frac{175}{9} + \frac{625}{9}e^{-0.72}\right). \end{aligned}$$

Therefore

$$\begin{aligned} -15 + 15e^{0.48} &= \frac{c}{m}\left(-\frac{175}{9} + \frac{625}{9}e^{-0.72}\right) \\ \frac{c}{m} &= \frac{-15 + 15e^{0.48}}{-175/9 + 625e^{-0.72}/9} \\ &= \frac{-135 + 135e^{0.48}}{-175 + 625e^{-0.72}} \\ &= \frac{-27 + 27e^{0.48}}{-35 + 125e^{-0.72}} \\ &= \frac{27e^{0.48} - 27}{125e^{-0.72} - 35}. \end{aligned}$$

43 h)

For $0 \le t \le 6$:

$$\begin{aligned} v &= \frac{c}{m}\left(\frac{25}{3}t - \frac{625}{9} + \frac{625}{9}e^{-0.12t}\right), \\ s &= \frac{c}{m}\int_0^6 \left(\frac{25}{3}t - \frac{625}{9} + \frac{625}{9}e^{-0.12t}\right) dt \\ &= \frac{c}{m}\left[\frac{25}{6}t^2 - \frac{625}{9}t - \frac{15625}{27}e^{-0.12t}\right]_0^6 \\ &= \frac{c}{m}\left(\left(\frac{25}{6}(6)^2 - \frac{625}{9}(6) - \frac{15625}{27}e^{-0.12(6)}\right) - \left(0 - 0 - \frac{15625}{27}\right)\right) \\ &= \frac{c}{m}\left(\left(-\frac{800}{3} - \frac{15625}{27}e^{-0.72}\right) + \frac{15625}{27}\right) \\ &= \frac{c}{m}(30.35170373\ldots) \\ &= 19.53528415\ldots. \end{aligned}$$

For $6 \le t \le 10$:

$$v = -15 + 15e^{1.2-0.12t}$$

$$
\begin{aligned}
s &= \int_6^{10} (-15 + 15e^{1.2-0.12t})\,dt \\
&= [-15t - 125e^{1.2-0.12t}]_6^{10} \\
&= (-15(10) - 125e^{1.2-0.12(10)}) - (-15(6) - 125e^{1.2-0.12(6)}) \\
&= (-150 - 125e^{0}) - (-90 - 125e^{0.48}) \\
&= -275 + 90 + 125e^{0.48} \\
&= -185 + 125e^{0.48} \\
&= 17.00930027\ldots.
\end{aligned}
$$

Total distance $= 19.53528415\ldots + 17.00930027\ldots + \cdots = 37$ m to the nearest metre.

44 a)

The rate of increase being proportional to the difference between limiting speed and current speed means

$$\frac{dv}{dt} = k(V - v) \quad \Rightarrow \quad \frac{dv}{dt} + kv = kV.$$

44 b)

Given $v = V(1 - e^{-kt})$,

$$
\begin{aligned}
v &= V - Ve^{-kt} \\
\frac{dv}{dt} &= Vke^{-kt} \\
k(V - v) &= k(V - (V - Ve^{-kt})) = k(V - V + Ve^{-kt}) = Vke^{-kt} \\
\therefore \frac{dv}{dt} &= k(V - v).
\end{aligned}
$$

So it satisfies the differential equation. At $t = 0, v = 0$:

$$0 = V(1 - e^{-k(0)}) = V(1 - 1) = 0,$$

so it satisfies the initial condition. As $t \to \infty, e^{-kt} \to 0$, so $v \to V$, satisfying the long-term behaviour.

44 c)

The equation is

$$\frac{dv}{dt} + \frac{2t}{1 + t^2}v = \frac{2Vt}{1 + t^2} + F(t).$$

$$
\begin{aligned}
\text{IF} &= e^{\int \frac{2t}{1+t^2}\,dt} = e^{\ln|1+t^2|} = |1 + t^2| \\
\text{Let IF} &= 1 + t^2.
\end{aligned}
$$

44 d)

When $F(t) = 0$:

$$
\begin{aligned}
\frac{dv}{dt} + \frac{2t}{1 + t^2}v &= \frac{2Vt}{1 + t^2} \\
(1 + t^2)\frac{dv}{dt} + \frac{2t}{1 + t^2}v(1 + t^2) &= \frac{2Vt}{1 + t^2}(1 + t^2)
\end{aligned}
$$

$$
\begin{aligned}
(1+t^2)\frac{dv}{dt} + 2tv &= 2Vt \\
\frac{d}{dt}((1+t^2)v) &= 2Vt \\
(1+t^2)v &= Vt^2 + c \\
v &= \frac{Vt^2+c}{1+t^2}.
\end{aligned}
$$

Substitute $t = 0, v = 0$:

$$
0 = \frac{V(0)^2 + c}{1+(0)^2} \Rightarrow c = 0, \quad \therefore v = \frac{Vt^2}{1+t^2}.
$$

44 e)

Let $v = \frac{1}{2}V$:

$$
\begin{aligned}
\frac{1}{2}V &= \frac{Vt^2}{1+t^2} \\
\frac{1}{2} &= \frac{t^2}{1+t^2} \\
1+t^2 &= 2t^2 \\
1 &= t^2 \\
t &= 1.
\end{aligned}
$$

So it takes 1 hour = 60 minutes to reach half the limiting value.

44 f)

When $F(t) = e^{-t}$:

$$
\begin{aligned}
\frac{dv}{dt} + \frac{2t}{1+t^2}v &= \frac{2Vt}{1+t^2} + e^{-t} \\
(1+t^2)\frac{dv}{dt} + \frac{2t}{1+t^2}v(1+t^2) &= \frac{2Vt}{1+t^2}(1+t^2) + (1+t^2)e^{-t} \\
(1+t^2)\frac{dv}{dt} + 2tv &= 2Vt + (1+t^2)e^{-t} \\
\frac{d}{dt}((1+t^2)v) &= 2Vt + (1+t^2)e^{-t}.
\end{aligned}
$$

Let $I = \int (1+x^2)e^{-x}\,dx$. By parts, $u = 1+x^2,\ dv/dx = e^{-x},\ du/dx = 2x,\ v = -e^{-x}$:

$$
I = -(1+x^2)e^{-x} + \int 2xe^{-x}\,dx.
$$

Use integration by parts again, $u = 2x,\ dv/dx = e^{-x},\ du/dx = 2,\ v = -e^{-x}$:

$$
\begin{aligned}
I &= -(1+x^2)e^{-x} - 2xe^{-x} + \int 2e^{-x}\,dx \\
&= -(1+x^2)e^{-x} - 2xe^{-x} - 2e^{-x} + k_1.
\end{aligned}
$$

Thus

$$
\begin{aligned}
(1+t^2)v &= Vt^2 - (1+t^2)e^{-t} - 2te^{-t} - 2e^{-t} + k \\
v &= \frac{Vt^2 + k - e^{-t}(t^2+2t+3)}{1+t^2}.
\end{aligned}
$$

Substitute $t = 0, v = 0$:

$$\begin{aligned} 0 &= \frac{V(0)^2 + k - e^0(0^2 + 2(0) + 3)}{1 + (0)^2} \\ 0 &= k - 3 \\ k &= 3 \\ v &= \frac{Vt^2 + 3 - e^{-t}(t^2 + 2t + 3)}{1 + t^2}. \end{aligned}$$

44 g)

From (e), the first model takes 60 minutes to reach half-speed, not close to 34 minutes. Convert 34 minutes:

$$34/60 = 17/30 \text{ hours.}$$

For $V = 12, t = 17/30$, the first model gives

$$v = \frac{12(17/30)^2}{1 + (17/30)^2} = 2.9167\ldots.$$

The second model gives

$$v = \frac{12(17/30)^2 + 3 - e^{-17/30}((17/30)^2 + 2(17/30) + 3)}{1 + (17/30)^2} = 3.2743\ldots.$$

The second value is closer to 6, half the limiting speed, so the second model is more consistent with this information.

45 a)

$$t\frac{dS}{dt} = S - \frac{3t^2}{\sqrt{1 + t^2}}.$$

Rearrange into the form $dS/dt + P(t)S = Q(t)$:

$$\begin{aligned} t\frac{dS}{dt} - S &= -\frac{3t^2}{\sqrt{1 + t^2}} \\ \frac{dS}{dt} - \frac{1}{t}S &= -\frac{3t}{\sqrt{1 + t^2}} \\ \text{IF} &= e^{-\int \frac{1}{t}\,dt} = e^{-\ln|t|} = e^{\ln|t^{-1}|} = |1/t| \\ \text{Let IF} &= \frac{1}{t} \\ \frac{1}{t}\frac{dS}{dt} - \frac{S}{t^2} &= -\frac{3}{\sqrt{1 + t^2}} \\ \frac{d}{dt}\left(\frac{S}{t}\right) &= -\frac{3}{\sqrt{1 + t^2}} \\ \frac{S}{t} &= -3\,\text{arsinh}\,t + c \\ S &= -3t\,\text{arsinh}\,t + ct. \end{aligned}$$

Substitute $S = 8, t = 2$:

$$\begin{aligned} 8 &= -3(2)\,\text{arsinh}\,2 + 2c \\ 8 &= -6\,\text{arsinh}\,2 + 2c \\ c &= 4 + 3\,\text{arsinh}\,2 \\ S &= -3t\,\text{arsinh}\,t + (4 + 3\,\text{arsinh}\,2)t. \end{aligned}$$

45 b)

Set $S = 0$:

$$\begin{aligned} -3t \operatorname{arsinh} t + (4 + 3 \operatorname{arsinh} 2)t &= 0 \\ (-3 \operatorname{arsinh} t + 4 + 3 \operatorname{arsinh} 2)t &= 0 \\ t = 0 \quad \text{or} \quad -3 \operatorname{arsinh} t + 4 + 3 \operatorname{arsinh} 2 &= 0 \\ \operatorname{arsinh} t &= \frac{4}{3} + \operatorname{arsinh} 2 \\ t &= \sinh\left(\frac{4}{3} + \operatorname{arsinh} 2\right) \text{ seconds.} \end{aligned}$$

45 c)

We want $dS/dt|_{t=0}$. From the DE,

$$\begin{aligned} t\frac{dS}{dt} &= S - \frac{3t^2}{\sqrt{1+t^2}} \\ \frac{dS}{dt} &= \frac{S}{t} - \frac{3t}{\sqrt{1+t^2}} \\ &= \frac{-3t \operatorname{arsinh} t + (4 + 3 \operatorname{arsinh} 2)t}{t} - \frac{3t}{\sqrt{1+t^2}} \\ &= -3 \operatorname{arsinh} t + (4 + 3 \operatorname{arsinh} 2) - \frac{3t}{\sqrt{1+t^2}}. \end{aligned}$$

Therefore

$$\begin{aligned} \left.\frac{dS}{dt}\right|_{t=0} &= -3 \operatorname{arsinh} 0 + (4 + 3 \operatorname{arsinh} 2) - \frac{3(0)}{\sqrt{1+0^2}} \\ &= 4 + 3 \operatorname{arsinh} 2. \end{aligned}$$

46 a)

Acceleration is $a = dv/dt$, mass is 2 kg, and $F = ma$:

$$\begin{aligned} \frac{8}{t^2+2t+2} - \frac{2v}{(t+1)(t^2+2t+2)} &= 2\frac{dv}{dt} \\ \frac{dv}{dt} &= \frac{4}{t^2+2t+2} - \frac{v}{(t+1)(t^2+2t+2)} \\ \frac{dv}{dt} + \frac{v}{(t+1)(t^2+2t+2)} &= \frac{4}{t^2+2t+2}. \end{aligned}$$

46 b)

$$\frac{dv}{dt} + \frac{v}{(t+1)(t^2+2t+2)} = \frac{4}{t^2+2t+2}.$$

Integrating factor:

$$e^{\int \frac{1}{(t+1)(t^2+2t+2)}\,dt}.$$

Partial fractions:

$$\frac{1}{(t+1)(t^2+2t+2)} \equiv \frac{A}{t+1} + \frac{Bt+C}{t^2+2t+2}$$

$$1 \equiv A(t^2+2t+2) + (Bt+C)(t+1).$$

Let $t=-1$: $1=A$, so $A=1$. Let $t=0$: $1=2+C$, so $C=-1$. Let $t=1$: $1=5+2(B-1)$, so $B-1=-2$, $B=-1$. Hence

$$\frac{1}{(t+1)(t^2+2t+2)} \equiv \frac{1}{t+1} - \frac{t+1}{t^2+2t+2}.$$

Therefore the integrating factor is

$$\begin{aligned} e^{\int (\frac{1}{t+1} - \frac{t+1}{t^2+2t+2})\,dt} &= e^{\ln|t+1| - \frac{1}{2}\ln|t^2+2t+2|} \\ &= e^{\ln|\frac{t+1}{(t^2+2t+2)^{1/2}}|} \\ &= \left|\frac{t+1}{(t^2+2t+2)^{1/2}}\right|. \end{aligned}$$

Let the integrating factor be $\frac{t+1}{\sqrt{t^2+2t+2}}$. Multiplying the DE through:

$$\begin{aligned} \frac{t+1}{\sqrt{t^2+2t+2}}\frac{dv}{dt} + \frac{v}{(t+1)(t^2+2t+2)}\frac{t+1}{\sqrt{t^2+2t+2}} &= \frac{4}{t^2+2t+2}\frac{t+1}{\sqrt{t^2+2t+2}} \\ \frac{t+1}{\sqrt{t^2+2t+2}}\frac{dv}{dt} + \frac{v}{(t^2+2t+2)^{3/2}} &= \frac{4(t+1)}{(t^2+2t+2)^{3/2}} \\ \frac{d}{dt}\left(\frac{t+1}{\sqrt{t^2+2t+2}}v\right) &= \frac{4(t+1)}{(t^2+2t+2)^{3/2}}. \end{aligned}$$

Use substitution $u=t^2+2t+2$, so $du/dt = 2t+2$ and $dt = du/[2(t+1)]$:

$$\begin{aligned} \int \frac{4(t+1)}{(t^2+2t+2)^{3/2}}\,dt &= \int \frac{4(t+1)}{u^{3/2}}\frac{1}{2(t+1)}\,du \\ &= \int 2u^{-3/2}\,du \\ &= -4u^{-1/2} + c \\ &= -4(t^2+2t+2)^{-1/2} + c. \end{aligned}$$

Thus

$$\begin{aligned} \frac{t+1}{\sqrt{t^2+2t+2}}v &= -4(t^2+2t+2)^{-1/2} + c \\ v &= -\frac{4}{t+1} + \frac{c\sqrt{t^2+2t+2}}{t+1} \\ &= \frac{c\sqrt{t^2+2t+2}-4}{t+1}. \end{aligned}$$

Substitute $v=-2, t=0$:

$$\begin{aligned} -2 &= \frac{c\sqrt{0^2+2(0)+2}-4}{0+1} \\ -2 &= c\sqrt{2}-4 \\ c &= \frac{2}{\sqrt{2}} = \sqrt{2}. \end{aligned}$$

Hence

$$v = \frac{\sqrt{2}\sqrt{t^2+2t+2}-4}{t+1}$$

$$= \frac{\sqrt{2t^2+4t+4}-4}{t+1}.$$

Stationary means $v = 0$:

$$\begin{aligned}
0 &= \frac{\sqrt{2t^2+4t+4}-4}{t+1} \\
\sqrt{2t^2+4t+4} &= 4 \\
2t^2+4t+4 &= 16 \\
2t^2+4t-12 &= 0 \\
t^2+2t-6 &= 0 \\
(t+1)^2-1-6 &= 0 \\
(t+1)^2 &= 7 \\
t &= -1 \pm \sqrt{7}.
\end{aligned}$$

We can disregard $t = -1-\sqrt{7}$ as $t \geq 0$. Check $-1+\sqrt{7} \approx 1.64 < 2$, so it is in the required domain. The first time the particle is stationary is $t = -1+\sqrt{7}$ seconds.